INTERNSHIP REPORT FOR DYNAMIC PLAYER TRANSFER VALUE PREDICTION USING AI AND MULTI-SOURCE DATA

PRESENTED BY:

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☑ Week 1 Summary – Data Understanding & Requirement Analysis:

During Week 1, the primary focus was on understanding the project objectives and the scope of player transfer value prediction. I collected all required datasets, including Transfermarkt player statistics and FIFA sentiment data, and studied the key features influencing market value. The week involved initial problem framing, requirement analysis, and defining expected deliverables. Basic exploratory data analysis was also performed to identify missing values and inconsistencies. A structured plan for the upcoming milestones was finalized. Overall, this week provided clarity and foundation for the full project pipeline.

☑ Week 2 Summary – Data Cleaning & Preprocessing:

Week 2 focused entirely on preparing the raw datasets for modeling. Several cleaning operations such as handling missing values, standardizing column names, converting data types, and removing duplicates were performed. Text-based columns were encoded, and numerical fields such as market values and statistics were normalized. Sentiment dataset cleaning and merging were also completed. The week ended with the successful creation of a unified, well-structured dataset ready for feature engineering. This ensured data consistency across all future model training steps.

☑ Week 3 Summary – Advanced Feature Engineering & Sentiment Integration:

In Week 3, deeper feature engineering techniques were applied to enhance model learning. New features such as performance ratios, peak-value comparison, value-to-age trends, and position-based embeddings were created. Sentiment analysis scores were integrated with the player dataset to capture public perception. Additional transformations like scaling, temporal grouping, and categorical encoding were implemented. All engineered features were tested for correlation with market value. By the end, a final feature matrix combining performance, sentiment, and injury data was successfully prepared.

☑ Week 4 Summary – Baseline ML Model Development:

Week 4 involved building baseline machine learning models for transfer value prediction. Algorithms such as Random Forest and LightGBM were trained on the engineered feature set. Model evaluation was performed using RMSE and MAE metrics, and initial predictions were generated. Feature importance analysis helped identify the top predictors affecting player value. Visualization of errors and distributions was completed. Week 4 established a baseline benchmark to compare with advanced sequence models in later weeks.

☑ Week 5 Summary – LSTM Time-Series Modeling:

During Week 5, the focus shifted to deep learning using LSTM networks. Univariate and multivariate LSTM models were developed to predict future transfer values based on historical patterns. Encoder-decoder architectures were also implemented for multi-step forecasting. Synthetic time-series generation was

used where historical data was limited. Model performance was evaluated with loss curves, RMSE, and MAE metrics. The week concluded with trained LSTM models capable of forecasting future market trends.

☑ Week 6 Summary – Ensemble Model Development:

Week 6 emphasized building ensemble models to improve prediction accuracy. XGBoost and LightGBM models were trained using both raw features and LSTM-generated predictions. A stacked ensemble model combining multiple algorithms was created and tested. Feature importance and SHAP explainability plots were generated to analyze model decisions. Ensemble evaluation showed significant performance improvement. The week delivered fully functional boosted models and ensemble outputs.

☑ Week 7 Summary – Model Tuning & Final Evaluation:

Week 7 was dedicated to optimization and validation. Hyperparameter tuning was performed for XGBoost, LightGBM, and LSTM models using RandomizedSearch and grid-based approaches. Tuned models achieved significantly lower errors compared to baseline versions. Cross-validation ensured strong model generalization. Final evaluation metrics (RMSE, MAE, R^2) were recorded and compared across all models. The week concluded with the selection of the best-performing ensemble model.

☑ Week 8 Summary – Visualization, Reporting & Final Output:

In Week 8, the final model outputs were compiled and visualized. Interactive graphs for player trends, predictions, and error analysis were developed. A complete project report was prepared, documenting methodology, implementation, results, and insights. A UI design was also initiated for player-wise prediction. The final presentation deck summarizing the entire project was completed. Week 8 wrapped up the full pipeline and delivered the final polished outcome of the internship.